

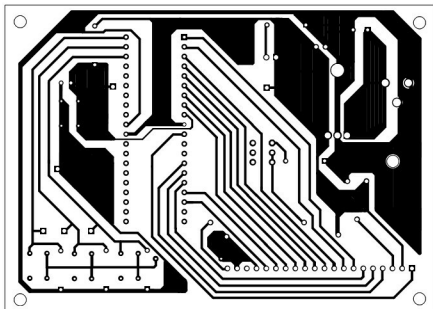


Fig. 4: An actual size, single-side PCB for the Space Invaders hand video game

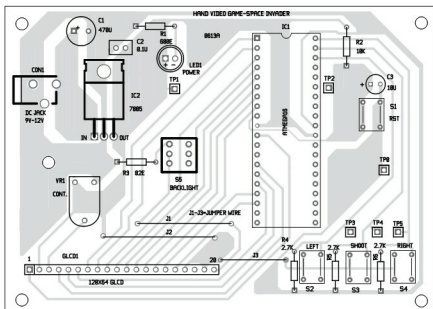


Fig. 5: Component layout for the PCB

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on www.efymag.com website.

tend to put more pressure on processors and are very prone to noise and other disturbances.

The last part is the main loop, which uses conditioned loops and statements determining the logic be-

hind which the game works. The detail about the code is commented in the code itself.

Construction and testing

An actual size, single-side PCB for the Space Invaders video game circuit is shown in Fig. 4 and its component layout in Fig. 5.

First, compile the code using WINAVR and burn the generated hex file into the MCU using a suitable programmer. Assemble the components



Fig. 6: Screenshot of the game

on the PCB as shown in the circuit diagram. Use an IC base for the ATmega16 microcontroller. You can use a 9V/12V adaptor or any other suitable DC source to power the circuit.

The game works in a simple manner. When you switch on the circuit, the GLCD screen displays 'Space Invaders' with game score board 'Score' appearing on the right half of the screen. After exactly four seconds, time score board 'Time' appears on the screen near the game score board. On the left half, you will find the tank or spaceship displayed at the bottom, and five animated insects/invaders dancing on the top.

The rules are simple. To win the game, shoot down the dancing insects within 10 seconds. Else, you lose. A score board on the right displays the number of insects shot down. Refer the screenshot of the game in Fig. 6.

If there is any display problem while testing, first check proper power supply. Also press reset switch S1 once or twice. If the problem persists, check the various test points as per the table given.

Note that an external crystal is not used here. If you want to improve the refresh rate, you can change the fuse byte of ATmega16 to `lfuse=0xfef` and `hfuse=0x99` from the command prompt in Avrdude software and connect a 4MHz crystal. You can also change the fuse byte by selecting 'security and configuration bits...' option in PonyProg2000 software.

Enjoy building your own video game and play! ●

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